

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a)	val°	the converse of val , which reads digits as a number
(b)	$([\text{wrap}, \text{snoc}]^\circ \text{Digit}^{++} (-\times \text{Digit}(\text{val})) [\text{embed}, \text{op}])^\circ$	val undoes wrap or snoc , reads the front with val and combines by embed or op , so its converse runs that backwards
(c)	$(\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ - \times \text{Digit}(\text{val}) \text{op})^\circ$	undoing wrap or snoc first gives two cases: a decimal and its last digit read by val then op , or one digit read by embed
(d)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val})^\circ \text{snoc}$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then snoc ; or undo embed , then wrap
(e)	$\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$	undoing the read of the front while keeping the last digit is val° on the front alone

(6.1b)

$$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$$

op° splits a number into its quotient and remainder by 10

$$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$$

op° gives a pair with a nonzero first component exactly at the numbers with two or more digits

$$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$$

embed° gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

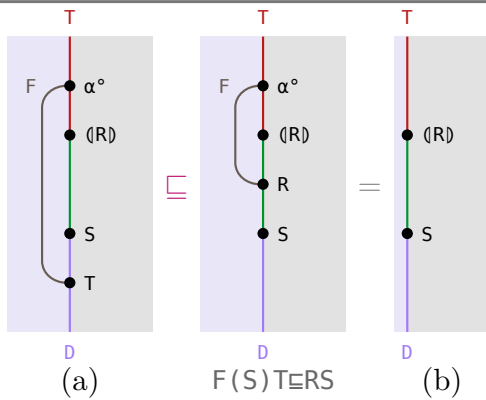
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow (R) \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq (R)$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

$$F(S)T \sqsubseteq RS \Rightarrow (T) \sqsubseteq (R)S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

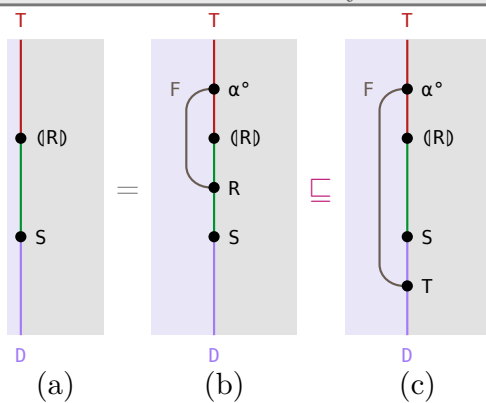


- (a) (6.2) at $X \triangleq (R)S$
 $F(R)S = F(R)F(S) \text{ — } \triangleq \text{ relator}$
 (b) $F((R))R = \alpha((R)), \alpha^\circ \alpha = 1 \text{ — } \triangleq \text{ reduce}$

$$RS \sqsubseteq F(S)T \Rightarrow (R)S \sqsubseteq (T)$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



- (a) (6.3) at $X \triangleq (R)S$
 (b) $\alpha((R)) = F((R))R, \alpha^\circ \alpha = 1 \text{ — } \triangleq \text{ reduce}$
 (c) $RS \sqsubseteq F(S)T$
 $F(R)F(S) = F(RS) \text{ — } \triangleq \text{ relator}$

§6.3. Hylomorphisms

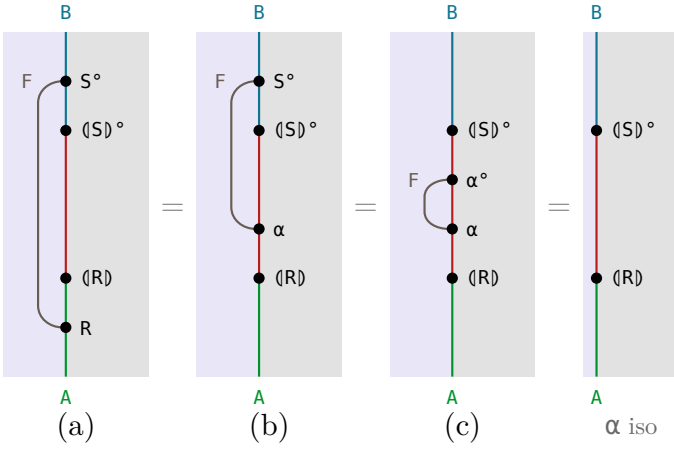
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

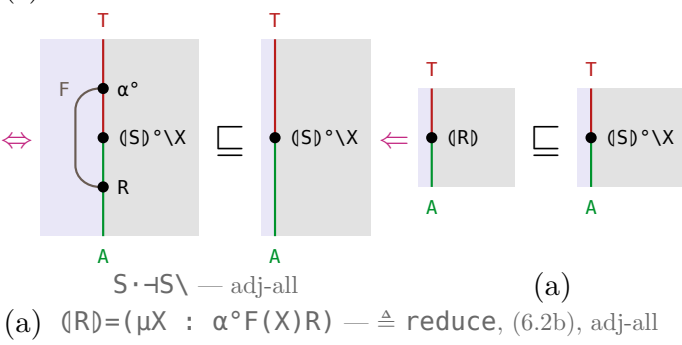
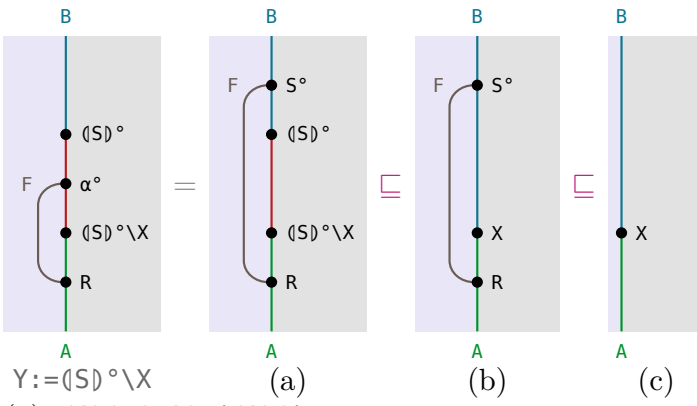
$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



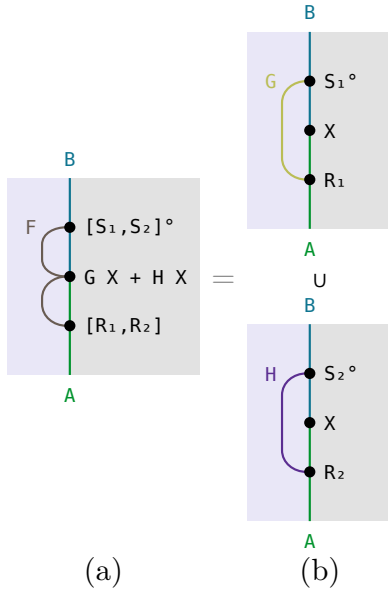
$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \circ ([R_1, R_2])^\circ) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

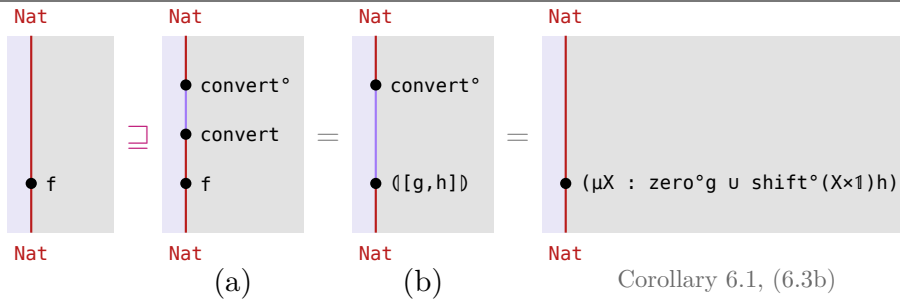


(a)	$[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$	the body of Theorem 6.2 at $F(X)=G(X)+H(X)$
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the $+$ stands for the codiagonals ∇°, ∇ , which are not drawn $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



Corollary 6.1, (6.3b)

- (a) convert simple
(b) fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of **a** to the power **n** and **d**

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

0 mod **b** is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

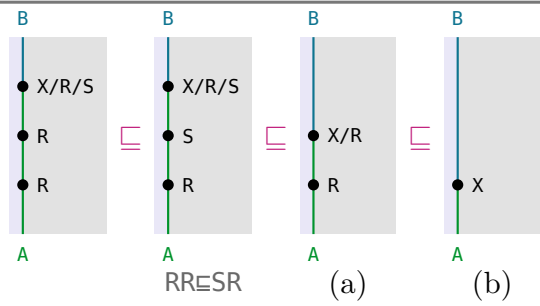
$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of **a mod b** and **d**

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $R \sqsubseteq SR$ for an inductive S



- (a) division by S
(b) division by R

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component

$$\text{inductive}(\text{Smember}) \wedge X = \text{SF}(X)R \wedge Y = \text{SF}(Y)R \Rightarrow X = Y$$

(6.5c)

when S **member** is inductive, two solutions X, Y of $X = \text{SF}(X)R$ are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge \text{SF}(X)R \sqsubseteq X \Rightarrow \text{Entire}(X)$$

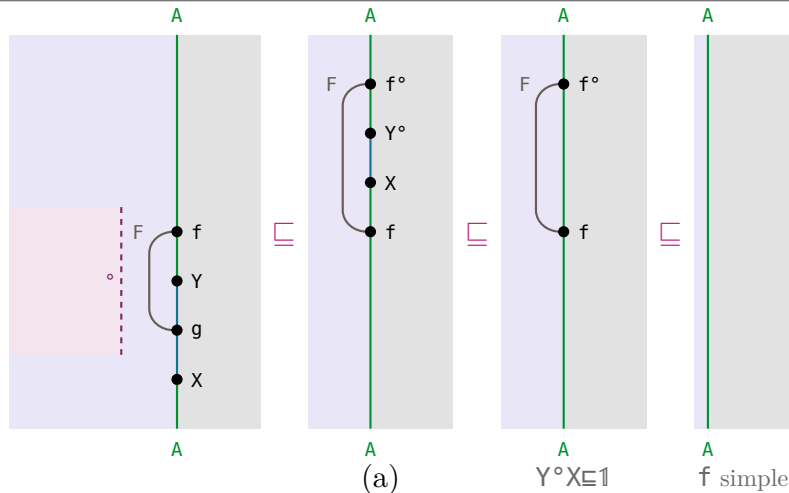
(6.5d)

when S **member** is inductive and S, R are entire, a pre-fixed point of $X \mapsto \text{SF}(X)R$ is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when g **member** is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

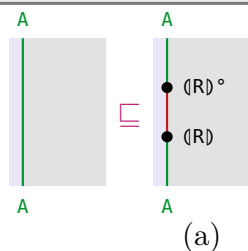


- (a) $X = gF(X)f$, g simple

$$\text{inductive}(R^\circ \text{member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective

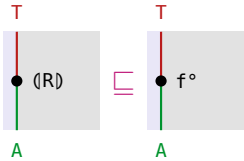


- (a) (6.5d) at $S = R^\circ$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

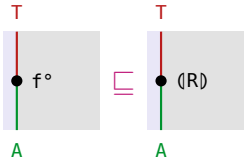
(6.5g)

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)

(a) shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$



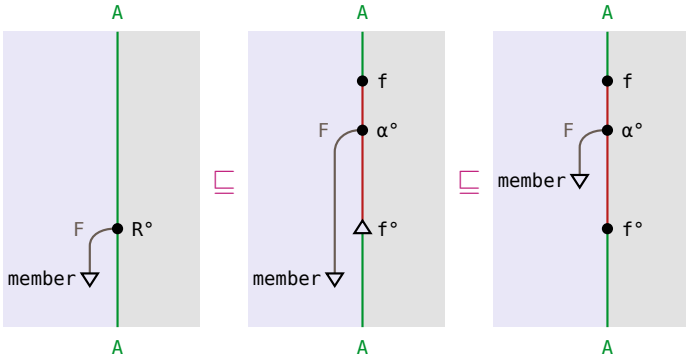
(a)

(a) (6.5f) and $\langle R \rangle f \sqsubseteq 1$

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)

(b)

(a) $Rf \sqsubseteq F(f)\alpha$, shunting

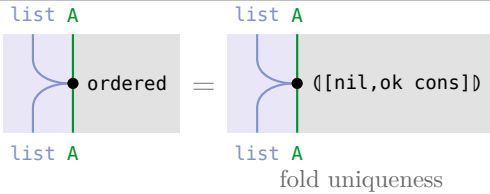
(b) member lax natural

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

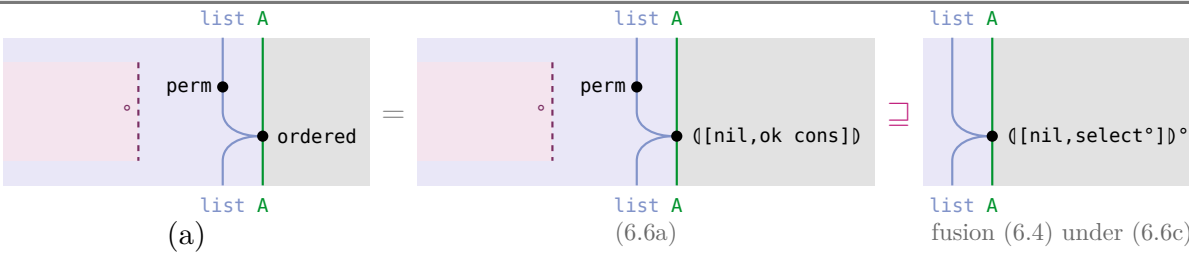


fold uniqueness

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it



(a)

(6.6a)

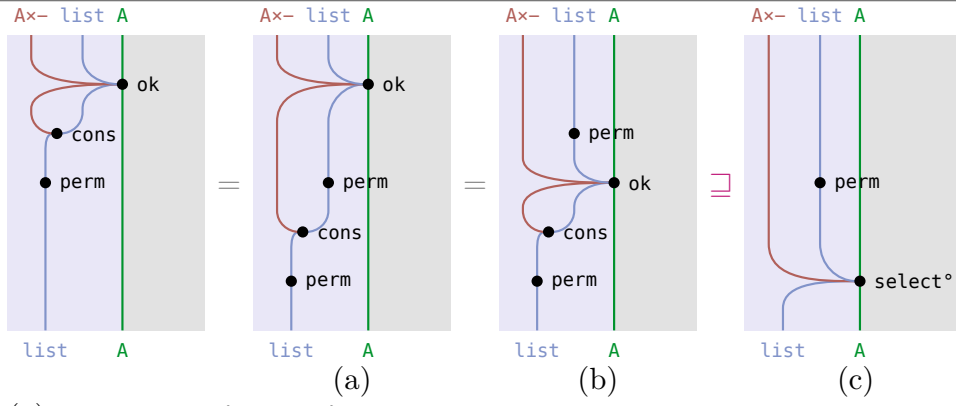
fusion (6.4) under (6.6c)

(a)

$\text{perm}^\circ = \text{perm}$, $\text{ordered}^\circ = \text{ordered}$

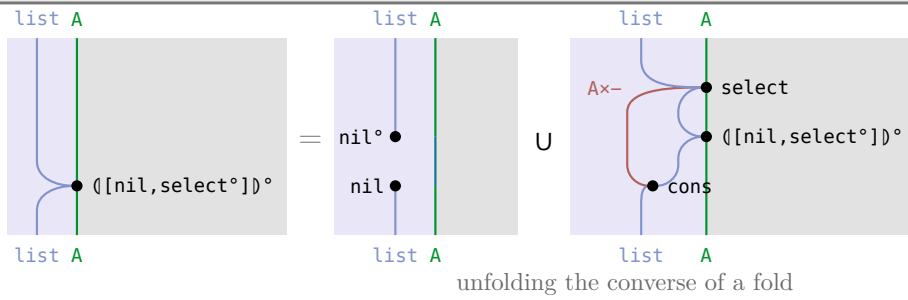
$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \vee \text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

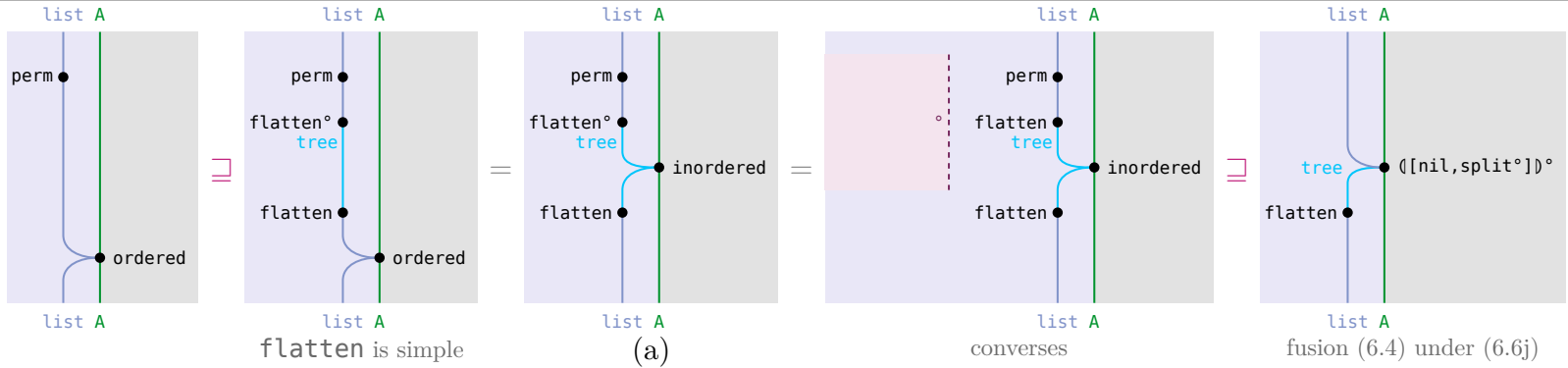
selection sort returns **[]** on **[]**, and otherwise selects **(a,y)**, sorts **y** and conses **a** back on



$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

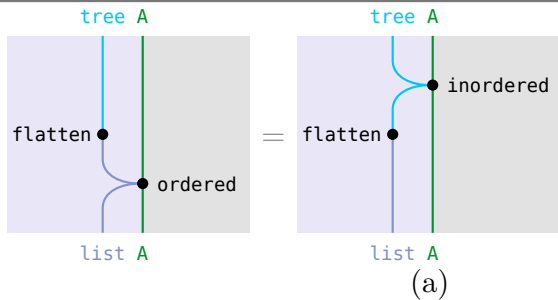
$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

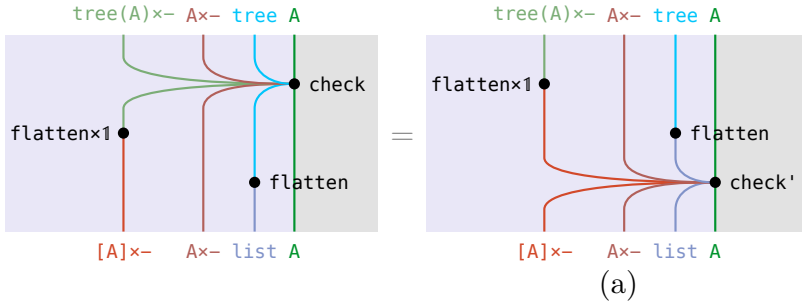
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



check (flatten×(1×flatten))=(flatten×(1×flatten))check'

(6.6g)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

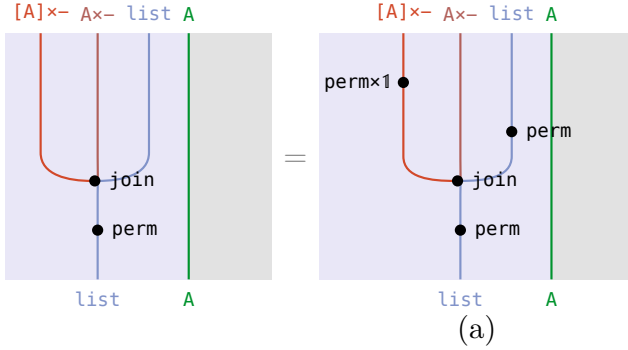


(a) $b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$

join perm=(perm×(1×perm))join perm

(6.6h)

permuting the list $x \mathrel{++} [a] \mathrel{++} y$ is the same as first permuting x and y and then permuting the joined list

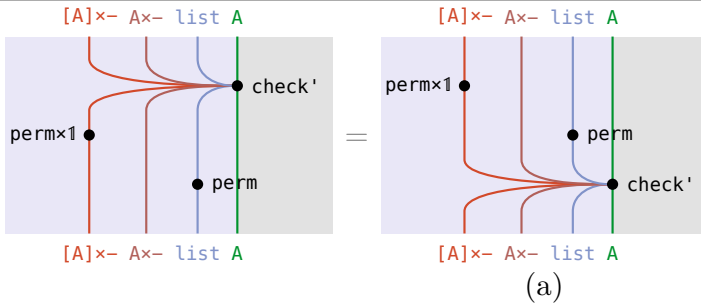


(a) **perm** is transitive and closed under $++$

check' (perm×(1×perm))=(perm×(1×perm))check'

(6.6i)

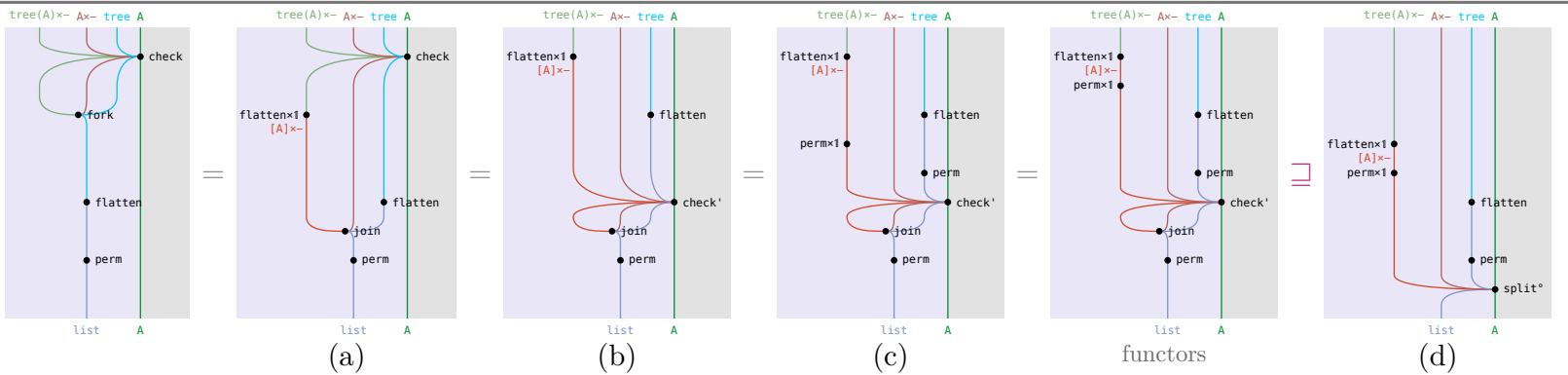
testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged



(a) $b \text{ inlist } x \iff b \text{ inlist } \text{perm}(x)$

split°⊆check' join perm $\implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \subseteq \text{check fork flatten perm}$ (6.6j)

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



(a) $\text{fork flatten} = F(\text{flatten})\text{join}$

(b) $\text{check } F(\text{flatten}) = F(\text{flatten})\text{check'}$

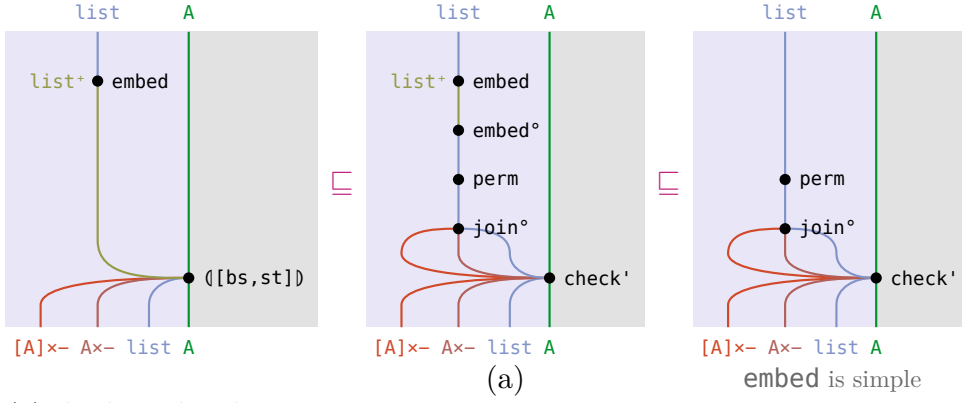
(c) $\text{join perm} = F(\text{perm})\text{join perm}, \text{check' } F(\text{perm}) = F(\text{perm})\text{check'}$

(d) $\text{split}^\circ \subseteq \text{check' join perm}$

$$\text{bs}\text{Ewrap perm join}^\circ\text{check}' \wedge (1 \times (\text{perm join}^\circ\text{check}')) \text{stEcons perm join}^\circ\text{check}' \Rightarrow \text{embed}([bs, st]) \text{E} \quad (6.6k)$$

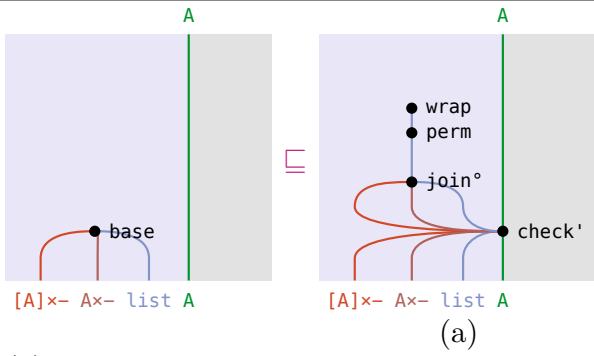
perm join[°]check'

turning a list into a non-empty list and folding it with any $[bs, st]$ meeting the two conditions below gives only **checked triples (x, a, y)** whose $\text{join } x \# [a] \# y$ is a permutation of the list



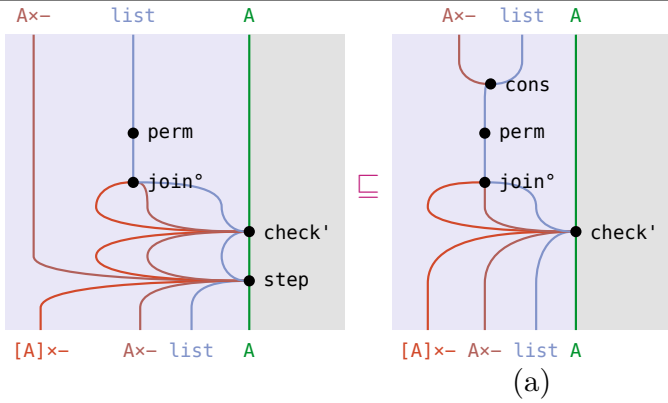
$$\text{base}\text{Ewrap perm join}^\circ\text{check}' \quad (6.6l)$$

base(a) is a **checked triple** whose **join** is a permutation of the one-element list $[a]$



$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ\text{check}')) \text{stepEcons perm join}^\circ\text{check}' \quad (6.6m)$$

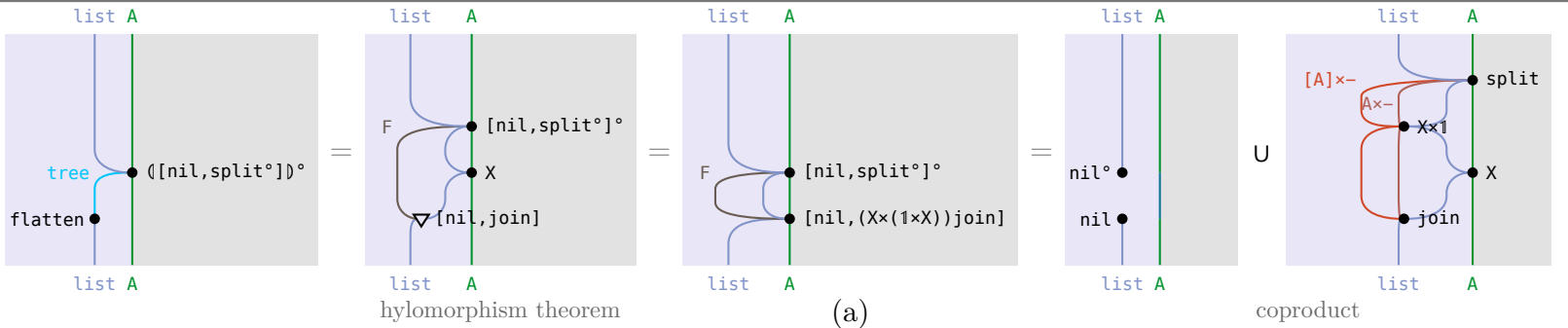
if (x, b, y) is a **checked triple** for a permutation of l , then $\text{step}(a, (x, b, y))$ is a **checked triple** for a permutation of $[a] \# l$



$$([nil, \text{split}^\circ])^\circ \text{flatten} = nil^\circ nil \cup \text{split} \quad (6.6n)$$

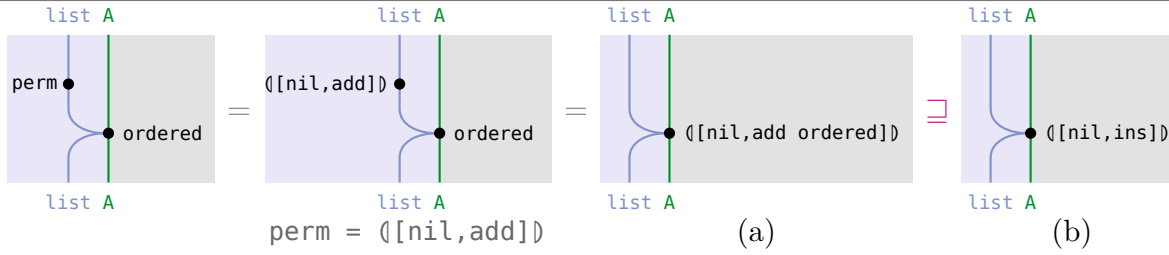
$$((([nil, \text{split}^\circ])^\circ \text{flatten}) \times (1 \times ([nil, \text{split}^\circ])^\circ \text{flatten})) \text{join}$$

quicksort returns $[]$ on $[]$, and otherwise splits into (x, a, y) , sorts x and y and joins them around a



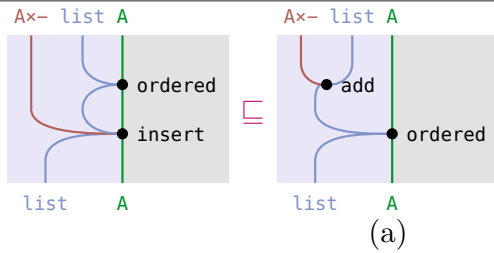
$$(1 \times \text{ordered}) \text{ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}]) \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



$$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{insert} \sqsubseteq \text{add ordered} \quad (6.6p)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(a) **insert** splices **a** in, and **R** transitive and total

§6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \sqsubseteq X \wedge XX \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^* \quad (6.7c)$$

R* is reflexive

	(a)	1	
$\sqsubseteq$	(b)	$1 \cup RR^*$	union
$=$	(c)	R^*	fixed point, Thm 6.1

$$R \sqsubseteq R^* \quad (6.7d)$$

R* contains **R**

	(a)	R	
$=$	(b)	R	identity
$\sqsubseteq$	(c)	RR^*	$1 \sqsubseteq R^* \Rightarrow (6.7c)$
$\sqsubseteq$	(d)	R^*	fixed point

$$R^* R^* \sqsubseteq R^* \quad (6.7e)$$

R* is transitive, because **R*/R*** is a prefixed point of $X \mapsto 1 \cup RX$

	(a)	$(1 \cup R(R^*/R^*))R^*$	
$=$	(b)	$R^* \cup R(R^*/R^*)R^*$	$(P \cup Q)T = PT \cup QT$ — composition distributes over $\cup$
$\sqsubseteq$	(c)	$R^* \cup RR^*$	$(S/S) \sqsubseteq S$
$\sqsubseteq$	(d)	R^*	$RR^* \sqsubseteq R^* \Rightarrow (6.7d)$

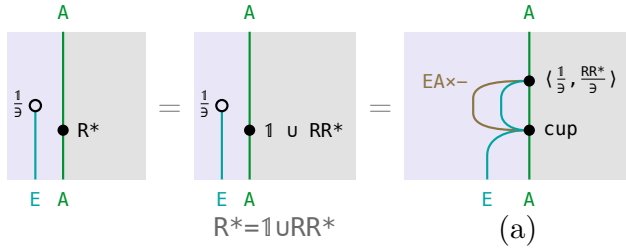
$$1 \sqsubseteq X \wedge XX \sqsubseteq X \wedge R \sqsubseteq X \Rightarrow R^* \sqsubseteq X \quad (6.7f)$$

a reflexive transitive **X** containing **R** is a prefixed point of $X \mapsto 1 \cup RX$, so it contains **R***

	(a)	$1 \cup RX$	
$\sqsubseteq$	(b)	$X \cup XX$	$1 \sqsubseteq X, R \sqsubseteq X$
$\sqsubseteq$	(c)	X	$XX \sqsubseteq X$

$$\frac{R^*}{S} = \left(\frac{1}{S}, \frac{RR^*}{S} \right) \text{cup} \quad (6.7g)$$

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors



$$(a) \ \Lambda(R \cup S) = \langle \Lambda(R), \Lambda(S) \rangle \text{cup}$$

$$R - \emptyset = R \quad (6.7h)$$

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y))) \quad (6.7i)$$

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

$$SR^* = (\mu X : S \cup XR) \quad (6.7j)$$

S followed by any number of R steps is the least X containing S and closed under a further R step

$$R^*S = (\mu X : S \cup RX)$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

$$\theta(\emptyset, S) = SR^* \quad (6.7k)$$

started with nothing found and S to explore, θ computes SR^*

	(a)	$\theta(\emptyset, S)$	
=	(b)	$\emptyset \cup (\mu X : S \cup (XR - \emptyset))$	definition of θ
=	(c)	$(\mu X : S \cup XR)$	$R - \emptyset = R$ — (6.7h)
=	(d)	SR^*	(6.7j)

$$\theta(P, \emptyset) = P \quad (6.7l)$$

with nothing left to explore, θ returns what it has found

	(a)	$\theta(P, \emptyset)$	
=	(b)	$P \cup (\mu X : \emptyset \cup (XR - P))$	definition of θ
=	(c)	$P \cup \emptyset$	$\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$
=	(d)	P	$P \cup \emptyset = P$